

BINGOL, Turkey

WMO: 172030

Lat: 38.885N

Lon: 40.501E

Elev: 3862

StdP: 12.76

Time Zone: 3.00 (E03)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	8.0	12.5	1.0	6.7	13.3	4.1	7.8	16.1	9.4	31.6	7.6	31.2	0.3	310	0.417

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	25.7	97.5	67.9	95.2	66.8	92.8	65.8	71.2	94.0	69.4	91.5	67.9	89.5	2.6	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.9	99.2	87.6	60.8	92.0	86.3	59.0	86.1	84.5	37.7	94.6	36.0	91.6	34.6	89.3	77.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.7	7.1	5.6	5.3	101.6	7.3	2.6	0.0	103.5	-4.3	105.1	-8.4	106.5	-13.7	108.4		
(5)			3.9	73.4	6.5	2.7	-0.8	75.3	-4.7	76.9	-8.3	78.5	-13.1	80.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	54.7	28.7	31.5	41.5	51.6	61.2	72.2	80.3	80.4	70.6	59.3	44.5	33.3		
	DBStd	19.18	7.65	7.76	7.74	7.04	6.24	5.16	4.49	3.54	5.38	6.53	7.32	8.40		
	HDD50	2239	659	518	280	63	4	0	0	0	0	10	186	519		
	HDD65	5155	1124	938	728	402	146	8	0	0	17	193	616	983		
	CDD50	3963	0	0	16	112	352	665	939	943	618	297	20	1		
	CDD65	1406	0	0	0	1	29	224	474	478	185	15	0	0		
	CDH74	16208	0	0	0	25	332	2368	5758	5645	1880	200	0	0		
	CDH80	7836	0	0	0	1	42	902	3096	3064	705	26	0	0		
	WSAvg	1.3	0.7	0.8	1.5	1.4	1.4	1.7	2.0	1.6	1.4	1.0	1.0	0.6		
	PrecAvg	33.3	4.5	4.9	4.7	4.2	3.0	0.9	0.2	0.2	0.6	2.4	3.7	4.3		
(15) Precipitation	PrecMax	43.3	14.0	10.9	9.0	9.8	8.5	2.7	1.0	1.0	3.1	8.1	8.1	9.2		
	PrecMin	21.8	0.4	1.9	1.4	1.3	0.4	0.1	0.0	0.0	0.0	0.2	0.1	0.2		
	PrecStd	5.4	2.7	2.0	1.9	2.0	1.7	0.6	0.2	0.2	0.7	2.0	2.1	2.5		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	47.4	53.6	67.3	77.2	84.1	93.2	101.3	99.9	93.6	83.0	67.7	56.9		
		MCWB	38.4	41.9	51.4	54.5	60.2	65.0	70.0	69.4	66.9	59.2	50.9	45.9		
	2%	DB	43.2	48.3	62.5	72.6	80.7	90.3	98.3	97.6	89.7	79.0	63.1	50.4		
		MCWB	36.6	40.1	48.4	53.1	59.3	65.0	68.1	67.9	64.2	57.5	49.5	41.2		
	5%	DB	40.9	44.8	58.5	68.3	77.9	88.0	95.6	95.5	87.1	75.6	59.7	46.4		
		MCWB	36.1	38.1	46.2	51.4	57.9	64.0	67.1	66.7	63.0	56.9	47.4	40.1		
	10%	DB	38.3	41.3	54.1	64.5	74.7	85.6	93.2	93.2	84.3	71.9	55.7	43.3		
		MCWB	34.4	36.4	44.5	50.3	57.0	63.1	66.3	65.7	61.8	55.7	45.2	38.7		
	0.4%	WB	41.2	43.8	53.1	57.0	63.1	69.3	73.9	73.5	69.1	62.0	52.6	46.7		
		MCDB	43.5	50.2	63.5	70.8	79.5	89.1	97.4	95.2	89.6	79.3	64.2	52.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	38.7	41.0	49.9	54.6	61.0	67.4	71.7	71.2	67.2	59.9	50.3	43.1		
		MCDB	41.4	46.8	60.1	68.2	77.6	87.3	94.7	94.1	86.9	75.4	60.3	47.7		
	5%	WB	36.7	39.1	47.6	52.8	59.3	65.8	70.0	69.5	65.1	58.2	48.5	41.2		
		MCDB	39.7	43.4	56.3	65.1	74.9	85.4	92.5	91.8	84.3	72.9	57.0	44.9		
	10%	WB	34.9	37.1	45.2	51.1	57.5	64.1	68.1	67.9	63.2	56.4	46.8	39.4		
		MCDB	37.7	41.0	53.1	62.3	72.2	83.5	90.3	89.8	82.0	69.9	54.0	42.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	12.3	13.5	16.3	18.2	21.5	24.5	25.7	26.8	26.5	22.3	18.3	13.5		
		MCDBR	14.8	17.1	22.9	25.0	26.2	26.2	27.8	28.9	29.7	28.0	25.0	18.1		
		MCWBR	9.7	10.7	12.3	12.1	11.8	12.6	12.9	13.4	14.7	13.6	13.6	11.1		
		MCDBR	11.4	14.6	19.4	21.4	23.7	25.4	26.4	26.9	27.6	25.0	22.1	15.5		
	5% WB	MCWBR	7.9	9.3	10.9	11.0	11.7	13.3	13.9	14.0	15.0	12.8	12.4	10.0		
(40) Clear-Sky Solar Irradiance	taub		0.305	0.340	0.383	0.438	0.403	0.356	0.359	0.362	0.330	0.371	0.323	0.303		
	taud		2.221	2.111	2.102	2.047	2.227	2.393	2.389	2.360	2.437	2.276	2.382	2.300		
	Ebn at Noon		271	275	275	266	279	292	289	286	291	262	265	264		
	Edh at Noon		32	41	46	52	44	38	37	37	33	35	27	28		
(44) All-Sky Solar Radiation	RadAvg		727	1044	1376	1611	2004	2480	2485	2215	1811	1192	824	628		
	RadStd		102	147	110	186	142	123	86	67	81	88	95	118		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (6 neighbors)	N/A	N/A	N/A	+1.22	-1.42	N/A	N/A	N/A	N/A	+61

Nomenclature: See separate page